

Institutional Trading and the Turn-of-the-Year Effect*

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Abstract

This study provides evidence that links institutional trading behavior directly to anomalous turn-of-the-year (TOY) return patterns of small stocks. We find that TOY trading patterns of institutions reflect strategies that are generally consistent with window-dressing and risk-shifting behaviors. Institutions sell more loser small stocks in the last quarter of the year, but buy more small stocks, both winners and losers, in the first quarter. Institutional buying (selling) of loser stocks at year-end weakens (strengthens) the TOY effect. In addition, buying of winner stocks after year-end causes a statistically significant, though weaker, effect.

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1 Introduction

Over the past two decades, the unusual return behavior of stocks, especially small stocks, around the turn of the year has not only intrigued the finance profession,¹ but also spawned much research attempting to explain this anomaly. Many explanations are offered, but the December tax-loss selling of stocks by individual investors and the year-end window dressing by institutional investors are the two popular, but competing, explanations given in the literature. Both the window-dressing and tax-loss trading strategies, however, largely produce similar turn-of-the-year stock return behavior. As a result, it is difficult for many studies to distinguish the two explanations and make definite conclusions, especially when they draw their inferences from observed return and volume patterns. Furthermore, while the two hypotheses are consistent with the abnormally large January returns generated by small stocks, whose value has fallen over the past year, neither of them seems to be consistent with the large January returns produced by small stocks, whose value has increased (e.g., Reinganum, 1983; Sias and Starks, 1997). In this study, we investigate whether institutional trading plays a role in the turn-of-the-year effect. More importantly, our analysis provides tests that link institutional trading behavior directly to the anomalous turn-of-the-year return patterns of small stocks.

Tax-loss selling is a strategy implemented by tax-motivated individual investors who have the desire to “dump” losing stocks at year-end in order to realize capital losses. As intuitively argued in the extant literature, this selling pressure would drive down stock prices at year-end; after the turn of the year, the pressure ceases, resulting in rising prices and abnormally large returns in January (the “January” effect). Earlier studies such as Rozeff and Kinney (1976), Givoly and Ovadia (1983), Keim (1983), Reinganum (1983), Roll (1983), and Lakonishok and Smidt (1984) have argued that tax-loss selling of loser stocks induces abnormal stock-return patterns at the turn of the year. The tax-loss selling

¹It has been well documented that stocks, especially small stocks, exhibit abnormally large returns during the first few days of January; this phenomenon is commonly referred to as the “turn-of-the-year” or “January” effect.

hypothesis is also consistent with the abnormally high volume in December (e.g., Dyl, 1977) and the individual investor buy- and sell-volume patterns around the turn of the year (e.g., Ritter, 1988).

On the other hand, window dressing is a strategy engaged primarily by non-tax-related motives of institutional investors. Popular wisdom is that institutional investors have incentives to “reshuffle” or “window dress” their portfolios in order to make their holdings look impressive in their reports (e.g., Haugen and Lakonishok, 1988; Lakonishok, Shleifer, Thaler, and Vishny, 1991, (LSTV); He, Ng, and Wang, 2002). LSTV show that institutional investors are more aggressive to get rid of losing stocks from their portfolios, especially just before the end of the year, when they must disclose their portfolio holdings. They therefore argue that this is evidence of window dressing. Haugen and Lakonishok offer this window dressing activity by institutions as a plausible explanation for the January effect. He, Ng, and Wang (2002) show that incentives affect institutions’ desires to window dress their portfolios.

In spite of this voluminous past research, none has performed direct tests of the tax-loss trading hypothesis vis-à-vis the window-dressing hypothesis. Several recent studies have attempted, though indirectly, to disentangle the two hypotheses. Poterba and Weisbenner (2001) and Grinblatt and Moskowitz (2003), respectively, analyze how early January returns and both December and January returns vary with changes in tax rules relating to capital gains. These tax regime changes should only affect the incentives for year-end tax-motivated trading by individual investors and not the incentives for window dressing by institutional investors. Sias and Starks (1997) employ cross-sectional differences of individual and institutional ownerships of stocks to examine whether tax-loss trading by individual investors, or window dressing by institutional investors, explains the turn-of-the-year effect. All these studies show patterns of stock returns to be consistent with the tax-loss selling hypothesis. Even though this recent evidence is more compelling, one is still unable to establish an explicit linkage between turn-of-the-year return anomalies and the trading activity of

individual investors versus institutional trading behavior.

Furthermore, neither the tax-loss selling hypothesis nor the window-dressing hypothesis is able to provide satisfactory explanations for the turn-of-the-year effect observed in winner small stocks. This therefore motivates us to explore the risk-shifting strategies by institutions as another plausible explanation for the January effect. Recent studies such as Brown, Harlow and Starks (1996) and Chevalier and Ellison (1997) find that mutual fund managers alter or manipulate the risk of their funds, in response to relative performance-driven compensation incentives. If institutions resort to risk-shifting strategies, as these studies argue, then the best time for them to load up riskier, typically smaller, stocks is immediately after they file their year-end portfolio holdings with the Securities and Exchange Commission (SEC), i.e., at the beginning of the new calendar year. It is possible that their buying activity of small stocks after the turn of the year is responsible for the turn-of-the-year effect in winner small stocks.

In this paper, we examine for evidence of institutional trading strategies that can induce the turn-of-the-year effect. Our work differs from most of the previous studies in that we employ detailed institutional equity holdings data that enable us to examine for evidence of a direct relation between institutional trading and turn-of-the-year returns anomalies. Our current study is partly motivated by the increasingly influential role of institutions in U.S. equity markets. Institutions owned more than 50 percent of the equity market share in the U.S. in 1998, as compared to 27.6 percent in 1980.² Previous research has also shown that institutions act in a concerted manner; they possess similar information and adopt similar trading strategies.³ It is therefore of interest to investigate whether their concerted trading activity affects turn-of-the-year return patterns. Our research reveals several findings that not only provide many important insights into the general investment behavior of institutions around the turn of the year, but also allow us to determine the driving forces behind the turn-of-the-year return patterns of small stocks.

²Gompers and Metrick (2001) document that the market value of institutional equity holdings had risen from \$375 billion in 1980 to \$3.98 trillion in 1996.

³See Wermers (1999) and Badrinath and Wanal (2002)

Our study provides new and interesting results. We find that the turn-of-the-year trading patterns of institutions reflect strategies that are generally consistent with window-dressing and risk-shifting behaviors. Institutions tend to sell more extreme loser small stocks in the last quarter of the year, but buy more small stocks, both winners and losers, in the subsequent quarter. We further provide evidence of the direct relation between institutional trading and turn-of-the-year return patterns of the small stocks that they trade in. When institutional selling pressure of extreme loser small stocks is high, the turn-of-the-year effect becomes stronger. On the other hand, when institutional buying pressure of loser small stocks is high, the turn-of-the-year effect becomes weaker. Thus, institutions buying extreme loser small stocks helps to weaken the turn-of-the-year effect, while selling strengthens the effect. In addition, institutions buying more small stocks at the beginning of the year causes a statistically significant, albeit weaker, January effect in winner stocks. Our results reveal that small stocks with no institutional ownership exhibit a significant turn-of-the-year effect, and this finding is clearly attributable to tax-loss selling by individual investors. Thus, in contrast to existing studies, we show that the existence of tax-loss selling by individuals does not preclude the role of institutional trading in the turn-of-the-year effect. Overall, the results imply that the turn-of-the-year effect in loser small stocks is not only driven by tax-loss selling by individuals, but also by institutional trading.

The paper is organized as follows. Section 2 describes the data and shows the trend of turn-of-the-year return patterns using our more recent sample. In Section 3 we examine the trading behavior of individual institutions around the turn of the year. In Section 4 we investigate the direct relation between aggregate institutional trading during the last and first quarters of the calendar year and turn-of-the-year prices of small stocks. The final section concludes.

2 Data Description

In this study we employ two data sets. The first set consists of institutional holdings data from CDA Investment Technologies, Inc., a service company engaged by the SEC to process and maintain 13(f) filings. The sample period spans from the first quarter of 1986 to the first quarter of 1999. The other set contains daily stock return data from the Center for Research in Security Prices (CRSP), and the same sample period is used in order to match the sample period for the holdings data. Below we describe the type of information that we use from the two data sets.

2.1 Institutional Stock Holdings

The institutional disclosure program mandated by Section 13(f) of the Securities Exchange Act of 1934 requires large institutional investment managers to file information about equity holdings under their investment discretion to the SEC. In 1978 the SEC adopted rule 13(f-1) and form 13F. The rule requires institutional managers exercising investment discretion over \$100 million in equity securities to file the information prescribed by form 13F with the SEC.⁴ The originally proposed rule required the form to be filed annually, but since 1979 the final version of the rule has required quarterly filings. The reporting requirements encompass various types of institutional managers such as banks, investment companies, pension funds, insurance companies, and brokerage houses. The reports contain information about equity holdings of reporting managers and the type of investment discretion they and their other investment managers exercise with respect to those securities.

Our study uses institutional holdings of common stocks traded on NYSE, AMEX, and NASDAQ. We exclude ADRs, foreign stocks, closed-end funds and REITs from the sample. As required by the SEC's form 13F, all common stock positions of these institutions greater than \$200,000 or 10,000 shares must be reported quarterly. As 13F institutions typically

⁴Throughout the paper, the terms "institutional investors", "institutional managers", and "institutions" are used synonymously to refer to 13(f) institutions or managers.

are “large” institutions, this requirement threshold should have little impact on institutions filing practice.⁵ The fact that institutions are not required to file such small holdings is more likely to bias our subsequent results towards finding no effect of institutional trading on small stocks. Nonetheless, we will address such issues when discussing the results below. The current study also includes both timely 13(f) filings and late-filings.⁶

Table 1 shows annual stock holding patterns of institutional investors across five size-based quintiles from 1986 to 1998. Within each quintile, it reports the total number of stocks, the proportion of stocks held by institutions, and the fraction of the market value of the stocks owned by institutions. The five size quintiles are constructed as follows. At the end of each September, we sort NYSE, AMEX, and NASDAQ stocks into five size quintiles according to the breakpoints of the market capitalization of NYSE stocks. These five size quintiles are used as the base portfolios throughout this study. This approach inevitably results in a large number of stocks falling within the smallest size quintile. From 1986 to 1998, the number of small stocks increases substantially from 3,627 to 4,622, as opposed to a relatively modest increase seen in the remaining four quintiles. In 1998, institutions have holdings in 94.4 percent of stocks in the smallest quintile and 99.8 percent of those in the largest size quintile. In terms of market capitalization, institutions own about 61 percent of the total value of large stocks and only 20 percent of the total value of small stocks. Consistent with Gompers and Metrick (2001), it is evident that institutions exhibit strong preference towards large stocks.

The table also dictates the growing dominance of institutional investors in U.S. equity markets over the 13-year sample period. While the increasing institutional ownership is

⁵Although institutions are not required to report their positions below the filing threshold, we find that many do disclose such holdings regularly.

⁶Under Rule 13f-1(a), any filing that is not made within 45 days after the last day of each quarter-end is considered a late filing. Under special circumstances, the SEC will grant late filings if they view this as necessary to protect the interest of investors. For more information, see “Commission Notice: Re: Section 13(f) Confidential Treatment Requests,” dated June 17, 1998; the article is available from <http://www.sec.gov/divisions/investment/guidance/13fpt2.htm>. For example, if an institution is engaged in an ongoing investment strategy that involves a large position change on the reporting date, it can request for a late filing. Nevertheless, our analysis showed that the results were not materially affected whether we included or excluded institutions with late-filings.

apparent across all size quintiles, this pattern is more dramatic in smaller stocks. In 1986, only 77 percent of the 3,627 smallest stocks have institutional ownership, but in 1998, this percentage increases to 94 percent of the 4,622 stocks. Correspondingly, in terms of market capitalization, institutional ownership of the smallest stocks rises from 9 percent to about 20 percent.

Throughout the sample period, close to 100 percent of large stocks have institutional ownership, but this high level of institutional ownership is not seen in smaller stocks. In the 1980's, on average 20 percent of small stocks have no institutional ownership. Although this percentage has fallen in the 1990's, on average more than 10 percent of small stocks still have no institutional holdings. In our later analysis, we specifically examine turn-of-the-year returns from this group of stocks not held by institutions and contrast them with those from stocks held by institutions.

2.2 Turn-of-the-Year Return Patterns

Here we re-establish the widely documented evidence on the turn-of-the-year effect for the sample period of December 1986 to January 1999. To be consistent with earlier studies, we report the return patterns for stocks across the five size quintiles. In contrast, however, within each size quintile, we also classify stocks into four winner/loser portfolios based on the stock's return performance from January to November of each year. We assign stocks to a winner (loser) group if the cumulative return of the stock over the past 11 months of the year is greater (less) than zero. Within the winner/loser groups, we further split them into two approximately equal groups. As a result, there are four performance quartiles in each size quintile. The bottom quartile consists of stocks with the smallest returns and is termed as the extreme loser quartile, while the top quartile contains stocks with the largest returns and is labelled the extreme winner quartile. We then calculate returns for the last ten trading days of December ('Dec') and the first ten trading days of January ('Jan') for

each stock.⁷ The 20-day period return measures the turn-of-the-year return.

It is necessary to point out that earlier studies have examined the impact of no trading and bid-ask spread on the January effect. For example, Keim (1989) has found that many of the smaller stocks have no trading for as long as four days after the turn of the year. We have also examined this no-trading issue using the daily volume of small stocks in our sample period. We find that the number of no-trading days of small stocks has declined steadily through the sample period. At the beginning of the sample period, about 20 percent of small stocks on average do not trade on a given day during the first ten trading days of the year, but this percentage drops to about 10 percent at the end of our sample period. For the ten-day trading period that we use in our study, only three percent of all of stocks in the smallest size quintile do not have any trading during the first ten days of 1986 and only one percent during the first ten days of 1999. Since our analysis focuses mainly on small stocks that institutions buy and sell, we find that institutions generally hold the larger stocks in the smallest size quintile. As we have confirmed by examining the data, the no-trading issue has virtually little impact on these stocks, because on average about 99.8 percent of all the small stocks held by institutions do have trading activity during the ten-day period. Thus, our use of a ten-day return window should help to mitigate the no-trading problem of small stocks.

Table 2 presents average returns for the last ten trading days of December (Dec), the first ten trading days of January (Jan), and the difference between January and December returns ('Diff(J, D)') for all stocks across size quintiles and also for extreme winner and loser stocks within each quintile. The results shown in the table are for the entire sample and for three approximately equal subperiods. Similar to prior studies, the turn-of-the-year effect is mainly a small-stock phenomenon. Throughout the 13-year sample period, and also for the three subperiods, only the smallest stocks exhibit significant turn-of-the-year effects. Small stocks have higher January returns and lower December returns, and the difference between

⁷Throughout the paper, December returns refer to returns for the last ten trading days of December and January returns refer to returns for the first ten trading days of January, unless otherwise specified.

January and December returns is positive and significantly different from zero at the five percent level. For the whole sample period, small stocks engender a 3.7 percent average return at the turn of the year, as compared to -1.3 percent for large stocks. Within small stocks, extreme loser stocks experience the largest turn-of-the-year effect: their $\text{Diff}(J, D)$ return varies from 5.9 percent in December 1986-January 1990 to 9.2 percent in December 1994-January 1999. This turn-of-the-year effect also exists for extreme winner small stocks, but the effect, while weaker, is not persistent across all subperiods. For instance, extreme winners exhibit no turn-of-the-year effect during the subperiod of December 1995 to January 1999.

In summary, the turn-of-the-year effect documented in our more recent sample period is consistent with the evidence in earlier periods.⁸ Similar to the existing results that are reported in the literature, the effect exists predominantly in small stocks and is more pronounced in loser than winner small stocks. Although the overall evidence on the effect is weaker in the late 1990's, it remains strong for small loser stocks. The results presented in this subsection certainly motivate our primary focus of only small stocks, those within the bottom size-formed quintile, in the remaining part of this paper, unless otherwise stated.

3 Institutional Trading Behavior Around the Turn of the Year

In this section, we study the investment strategies of institutional investors around the turn of the year. In particular, we examine whether such strategies suggest that institutions engage in window dressing towards the end of the year and in adjusting the risk of their portfolios at the beginning of the year.

Existing studies have provided evidence that groups of institutions such as mutual funds, pension funds, or money market funds window dress their portfolios, in response to mandatory public disclosures or plan sponsors' evaluations. But none of these studies actually

⁸See Schwert (2001) for more recent evidence on the turn-of-the-year effect.

examine whether such trading behavior directly contributes to the turn-of-the-year effect. Furthermore, while the window-dressing activity by institutions is a reasonable argument for the observed January effect in loser small stocks, the same argument does not seem applicable to winner small stocks. This therefore leads us to explore the risk-shifting strategy as another plausible explanation for the January effect.

In this section, we employ quarter-end equity holdings for individual institutions to examine their trading activity, particularly their buying and selling of small stocks during the first and last quarters of the year. Such adjustments of stock positions may reflect institutional motives of window dressing their holdings, or altering the risk of their portfolios around the turn of the year. The results of these tests will enable us to evaluate the potential impact of institutional trading on prices of small stocks.

3.1 Window Dressing by Institutions

This subsection looks at the trading behavior of institutional investors during *fourth quarter*. We employ an approach similar to LSTV to test whether institutions actively window dress their portfolios by getting rid of stocks that have experienced declining prices over the past year in order to present “decent” year-end reports. We therefore construct an institution’s selling (buying) activity in a given performance quartile relative to its overall selling (buying) activity.

Based on institutional equity holdings between the end of September and the end of December, we identify all the stocks within each performance quartile that institutions sell and buy during the last quarter of the calendar year.⁹ The construction of performance quartiles of stocks is detailed in Section 2.2 above. The relative selling activity of institution k in each performance quartile j during the fourth quarter q is measured by

$$\frac{SELL(j, q, k)/HOLD(j, q - 1, k)}{\sum_j SELL(j, q, k)/\sum_j HOLD(j, q - 1, k)}, \quad (1)$$

⁹LSTV document that the turnover ratio calculated using the quarterly change of stock holdings is typical for institutions, thus suggesting that net transactions provide a reasonably good proxy for institutional sales and purchases during the quarter.

where $SELL(j, q, k)$ is the dollar value of sales by institution k in the fourth quarter q and in performance quartile j , and $HOLD(j, q - 1, k)$ is the dollar value of holdings of these stocks at the end of the third quarter. Values of $SELL$ and $HOLD$ are calculated as the average of the beginning and end of quarter q prices. We refer to Equation (1) as an institution's 'sell' ratio. The numerator of Equation (1) is institution k 's sales in a performance quartile j relative to its holdings of stocks in the same quartile j at the end of the third quarter. The denominator is institution k 's total sales in this quarter relative to total holdings at the end of the third quarter. Note that the sell ratio accounts for the fact that institutions may be selling more in a given performance quartile during the fourth quarter, because they are selling more of stocks across all quartiles. Suppose an institution were to sell 30 percent of the extreme loser stocks it holds, but only 20 percent of its total holdings. Its sell ratio calculated based on (1) should be 1.5. Hence the institution sells 50 percent more of loser stocks than it does in other stocks.

Similarly, we measure the relative buying activity by institution k in the fourth quarter q in performance quartile j , as

$$\frac{BUY(j, q, k)/HOLD(j, q, k)}{\sum_j BUY(j, q, k)/\sum_j HOLD(j, q, k)}, \quad (2)$$

where $BUY(j, q, k)$ is the dollar value of purchases by institution k in the last quarter of the calendar year q and in performance quartile j . The 'buy' ratio, as measured by Equation (2), shows the proportion of an institution's purchases in performance quartile j during the fourth quarter relative to the proportion of institution k 's total purchases across all performance quartiles.

We calculate the ratios (1) and (2) separately for each institution and then average the respective ratios over all institutions within each performance quartile at a given fourth quarter of the year, and then average them over the years, yielding equal-weighted ratios. Correspondingly, we obtain their value-weighted counterparts by calculating (1) and (2) for each institution, average the respective ratios across all institutions at the given quarter based on the relative value-weight of their holdings, and then average them across the 13

years of sample. All t -statistics of the mean coefficients are calculated based on their time-series standard errors. This approach will help to control the effect of cross-sectional dependence on the statistics reported. Throughout this study, we use the same approach to compute all the statistics reported in the remaining tables.

Table 3 presents time-series averages of equal- and value-weighted sell and buy ratios for extreme winner (top performance quartile) and loser stocks (bottom performance quartile). It also contains the difference of the buy and sell ratios, with their t -ratios reflected in parentheses. Panel A reports those sorted from the universe of stocks, without controlling for size. The results from this panel will provide an overall picture of institutional buying and selling activities. Panel B highlights the buy and sell ratios for extreme loser and extreme winner stocks from the smallest size quintile.

Panel A shows that on average institutions sell more than buy extreme loser stocks. Specifically, they sell about 58 percent (sell ratio = 1.58) more loser stocks, but only buy about 35 percent more (buy ratio = 1.35). Their value-weighted averages are 68 percent and 62 percent, respectively. The difference of the buy and sell ratios is statistically significant at conventional levels. Institutions on average also transact actively in winner stocks during the last quarter of the year, although the ratios are substantially smaller than those for loser stocks. They buy about 10 percent (buy ratio = 1.10) more winner stocks, whereas they sell only about 6 percent more (sell ratio = 1.06). Panel B depicts similar patterns of equal- and value-weighted buy and sell ratios for small stocks. It reports that institutions are inclined to sell more than buy loser small stocks. They sell about 27 percent more extreme loser small stocks (sell ratio = 1.27), but buy only 6 percent (buy ratio = 1.06) more of the same. The difference in the buy and sell ratios is about 22 percent and is statistically significant at the five percent level. Conversely, they buy about 12 percent (buy ratio = 1.12) more extreme winner small stocks, but sell about 3 percent less (sell ratio = 0.97).

In general, institutions are inclined to sell more loser stocks and also to buy more winner stocks during the last quarter of the year, and this behavior is evident in all stocks, including

small stocks. The results suggest that institutions window dress their portfolios in advance of mandatory disclosures. It is worthwhile to point out that such trading strategies can also be interpreted to be consistent with momentum trading. But it is difficult to fully differentiate the two explanations based solely on the results from the fourth-quarter’s institutional trading patterns. If institutions were to employ momentum trading strategies, their trading patterns should systematically manifest such strategies throughout the calendar year. In the following subsection, our analysis of institutional trading patterns during the first quarter of the year, however, provides no strong evidence to support the strategy. Thus, we are confident that the trading activity documented here is more likely to reflect window dressing than momentum trading. This interpretation is also consistent with Badrinath and Wahal (2002), who use a different methodology and find no evidence of institutional momentum trading.

3.2 Risk Shifting by Institutions

As we discussed earlier, it is plausible that not only window dressing by institutions, but also their risk-shifting strategies contribute to the turn-of-the-year effect in small stocks. If institutions systematically shift portfolio risks, we should expect to find evidence of changing portfolio compositions in their stock holdings. In particular, we should observe more institutional buying of riskier small stocks than large stocks, right after the December disclosure. The buying pressure of small stocks by institutions, in general, will push up small stock prices in early January. This risk-manipulating strategy probably contributes to the observed January effect in winner small stocks. Therefore, in this subsection, we first examine whether institutions manipulate the risk of their portfolios by buying more small stocks than large stocks during the *first quarter* of the year. If institutions do buy small stocks, we next proceed to investigate specifically whether institutions show any preference for winner versus loser small stocks.

In Table 4, we report time-series averages of mean equal- and value-weighted buy and sell ratios for the previously-constructed smallest and largest size quintiles of stocks held by

all institutions in the sample. It also reports the differences in the buy and sell ratios within each size quintile and the differences of the ratios between the largest and smallest quintiles of stocks. These buy and sell ratios are computed in the same manner as Equations (1) and (2) above, with a couple of exceptions. Here purchases and sales of stocks by institutions occur during the first quarter of the year, and the buy (sell) ratio measures the buying (selling) intensity of an institution in each size-quintile during the first quarter relative to its overall buying (selling) intensity across all size quintiles. In general, the reported ratios show that institutions trade more actively in small stocks than in large stocks. They tend to buy and sell more small stocks than large stocks, and at the same time, they also tend to buy more than sell small stocks. The equal-weighted buy ratio for small stocks is 1.50, which is substantially higher than the buy ratio of 0.96 for large stocks. The sell ratios for the small and large stocks are 1.28 and 1.01, respectively. All the differences are statistically significant at conventional levels. The results based on value-weighted ratios, while stronger, are consistent with those based on equal-weighted.

Overall, the results show that institutions on average increase their net holdings of small stocks, while buying fewer large stocks, during the first quarter of the year. They buy 50 percent more of small stocks but sell only 28 percent more of the same. On the other hand, they buy 4 percent less of large stocks, but sell one percent more of the stocks. It seems evident that institutions change the portfolio compositions of their stock holdings during the first quarter of the year. This result of buying small stocks may reflect risk shifting, where institutions intentionally manipulate the risk characteristics of their portfolios as a means to enhance performance. Alternatively, it may suggest “bargain hunting”, where institutions buy more loser small stocks, because these stocks that have just experienced price declines in late December might be particularly good buys. If it is the latter, the value-driven, or bargain-hunting, strategy should not be regarded as risk shifting. In order to rule out the alternative explanation, we proceed to examine the institutional trading activity, in more detail, by analyzing their trading of winner and loser small stocks.

Table 5 presents time-series averages of equal- and value-weighted buy and sell ratios for extreme loser and winner small stocks. Generally, institutions exhibit no clear preference for loser or winner small stocks. Based on equal-weighted ratios, institutions appear to prefer more winner than loser small stocks during the first quarter of the year. The buy ratio for winner small stocks is 1.11 and for loser small stocks is only 1.02, and their corresponding sell ratios are 1.00 and 1.14. The value-weighted ratios, however, show less obvious preference for either type of stock by institutions. Both extreme loser and winner stocks have the same buy ratios of about 1.17; their sell ratios are 1.10 and 1.15. The buy and sell ratios for the two middle quartiles, not reported, are on average about 0.90. The general patterns of buy and sell ratios suggest that institutions are active in trading both extreme winner and loser stocks during the first quarter of the year, and the evidence indicates more buying than selling of these stocks. These trading strategies further corroborate our earlier finding that institutions switch the risk of their portfolios at the beginning of the year.

Taken together, the results of Tables 4 and 5 provide reinforcing evidence that the beginning-of-the-year increase in institutional holdings of small stocks is not purely driven by value-motivated strategies of buying loser small stocks. In contrast to that of the last quarter of the year, there is no evidence that institutions consistently buy winner stocks and, at the same time, sell loser stocks after the turn of the calendar year. These results further confirm that institutional trading behavior during the last quarter of the year, as we interpreted earlier, is more in accord with window dressing than momentum investing.

4 Institutional Trading and Turn-of-the-Year Return Patterns

Thus far, we have found evidence of institutional trading strategies that are generally consistent with window-dressing and risk-shifting behaviors. Specifically, institutions tend to sell more extreme loser small stocks in the last quarter of the year, but buy more small stocks, both winners and losers, in the subsequent quarter. It is essential that we proceed

to test whether such trading strategies indeed contribute to the abnormally large January returns in small stocks. In this section, we examine for evidence of a direct relation between institutional trading and the turn-of-the-year return patterns of the small stocks that institutions trade in.

4.1 Effects of Net Buying and Selling of Small Stocks

The existing finding that tax-loss trading by individual investors drives the turn-of-the-year effect in small stocks is indisputably strong and compelling. Thus, one would naturally expect that individual investors as a group are net sellers of loser stocks at the end of the year and institutions in aggregate are liquidity providers. However, our preceding evidence of window dressing by institutions suggests that institutions are also actively selling loser small stocks before the December disclosure. It is therefore of interest that we examine the net buying and selling of small stocks by institutions. The results of our analysis ought to reveal more evidence on the overall trading activities of institutions at the end of the year and the extent to which such activities affect turn-of-the-year return patterns of small stocks.

We sort stocks from the bottom size quintile (consisting of small stocks only) into three different groups based on institutional trading activity in the fourth quarter of the year: stocks that are bought by institutions in aggregate during the last quarter of the year (activity $\equiv$ ‘Net Buyer’); stocks that are sold by institutions in aggregate during the last quarter of the year (activity $\equiv$ ‘Net Seller’); and stocks with no institutional ownership (activity $\equiv$ ‘No Holding’). For stocks to be classified in the Net-Buyer group, the change of institutional holdings as of the beginning and end of the fourth quarter (i.e., the end of September and the end of December) must be positive, and this represents an increase in the institutional holdings of the stocks during the last quarter of the calendar year. For stocks in the Net-Seller group, the change of institutional holdings in the fourth quarter must be negative, and this reflects a decline in the institutional ownership of the stocks. Finally, for stocks in the No-Holding group, these stocks must not be owned by any institution during

the fourth quarter; institutional holdings as of the end of September and of December contains no share of the stocks. We repeat the same sorting procedure to form three similar groups based on their trading activity by institutions in the first quarter of the year.

This method of classifying small stocks will permit us to examine whether institutional trading of small stocks helps to accentuate or mitigate the turn-of-the-year effect. If institutions in aggregate sell loser stocks, their selling activity should intensify the January effect. Conversely, if they buy in aggregate, their buying activity should either weaken or eliminate the effect in these loser small stocks. If institutions buy (sell) small stocks in aggregate after the turn of the year, the buying (selling) activity should strengthen (weaken) the January effect in the stocks. In other words, Net-Seller and Net-Buyer stocks should produce different turn-of-the-year return patterns.

It is important to emphasize that any evidence of institutional trading affecting small stock prices does not necessarily imply no tax-loss selling by individual investors. The impact of tax-loss selling should be evident in loser small stocks, especially those with no institutional-holding. Any anomalous January return from the latter has to be purely driven by tax-loss selling.¹⁰ As a result, any differential turn-of-the-year return between No-holding stocks and Net-Seller or Net-Buyer stocks will reflect the relative importance of institutional trading in the turn-of-the-year effect.

Panels A and B of Table 6 report time-series averages of turn-of-the-year returns on extreme winner and loser small stocks from each activity portfolio, where trading activity occurs in the last and first quarters of the year, respectively.¹¹ The turn-of-the-year returns are measured using the returns during the ten days prior to the end of December (Dec) and the returns for the ten days after the turn of the year (Jan). The table also reports their difference (Diff(J, D)) together with its t -statistic and the total number of observations

¹⁰As we mentioned in Section 2.1 above, it is also likely that institutions do have some ownership of these stocks but just do not require to report them. Given that their holdings, if any, are generally small, it is reasonable to assume that the impact of institutional trading on these stocks would be inconsequential.

¹¹Note that although the results are qualitatively similar for the second and third quartile portfolios, the two reported extreme performance quartiles provide sharper contrasting differences between them.

for each activity portfolio during the entire sample period. However, it does not report those from the remaining four size quintiles, because there exists no strong evidence of turn-of-the-year effects.¹²

Panel A of Table 6 shows that the December return of loser stocks from the Net-Seller group is significantly lower than that of the Net-Buyer's by 2.5 percent within the ten-day window, and this return difference is statistically significant at the five percent level. The panel also reports 8,351 Net-Seller observations but only 6,035 Net-Buyer's, implying that institutions in aggregate are more aggressive in selling than buying loser stocks. These findings suggest that institutional selling helps push down, while buying elevates, small stock prices, during the last ten days of December. After the turn of the calendar year, stock prices in general bounce back, yielding unusually large January returns of 8.2 percent in Net-Buyer's and 9.9 percent in Net-Seller's during the first ten days of the month. Consequently, loser stocks that institutional investors have sold in aggregate during the last quarter of the year yield an average return differential ($\text{Diff}(J, D)$) of 10.4 percent, whereas those that they have bought produce an average $\text{Diff}(J, D)$ return of only 6.9 percent. The average $\text{Diff}(J, D)$ return between Net-Seller and Net-Buyer stocks of 3.6 percent is statistically significant at the five percent level.

Undoubtedly, loser small stocks exhibit a strong January effect, and the effect is significantly more pronounced for stocks from the Net-Seller portfolio than the Net-Buyer's. Panel A shows no evidence of a January effect in winner stocks from either group. The turn-of-the-year returns on extreme winner stocks that institutions have bought or sold, in aggregate, during the last quarter of the year are substantially lower, ranging from 0.4 percent (Net-Buyer's) to 1.7 percent (Net-Seller's). Although institutions seem more active in buying (Obs.=6,493) than selling (Obs.=4,242) winner stocks, they are certainly more active in buying and selling loser than winner stocks.

Panel B of the table shows some contrasting results from those of Panel A. The January

¹²Detailed results on these four size quintiles can be available upon request.

effect in loser stocks becomes weaker when institutions are actively selling the stocks in the first quarter of the year, but it becomes stronger when institutions are buying them. The turn-of-the-year returns for Net-Seller and Net-Buyer stocks are 7.1 percent and 8.2 percent, respectively, as compared to 10.4 percent and 6.9 percent reported in Panel A. As in Panel A, there is no turn-of-the-year effect in Net-Seller and Net-Buyer winner stocks. Perhaps this calls for a finer grouping of the buying and selling intensities of institutions, and the following subsection addresses this issue.

Based on the number of observations reported in Panels A and B, they show that institutions are more active in selling loser stocks in the fourth than the first quarter of the year, but more active in buying both loser and winner stocks in the first quarter than the fourth quarter. These findings further reinforce our earlier evidence that institutions engage in window dressing by getting rid of loser stocks right before the end of December disclosure. Immediately thereafter, they turn to buying more winner and loser small stocks, a strategy consistent with the above finding of risk-adjusting behavior of institutions.

Table 6 also reveals that No-Holding stocks yield fairly consistent turn-of-the-year returns. Loser stocks in the No-Holding portfolio produce a turn-of-the-year return of 9.6 percent in Panel A and of 9 percent in Panel B, while their winner counterparts generate returns of 2.7 percent and 2.5 percent, respectively. Given the size differentials between the stocks in the No-Holding group and those held by institutions,¹³ we make no attempt to directly compare the turn-of-the-year returns of No-Holding stocks with those generated by Net-Seller and Net-Buyer stocks. While it is worthwhile to note that the No-Holding stocks, whose market capitalization is the smallest, do not produce the largest turn-of-the-year effect, their statistically-significant results should be interpreted as consistent with the well documented tax-loss selling effect.

Overall, the results of this subsection offer several important insights into the turn-

¹³In the sample period, the market capitalization of small stocks with no institutional holding on average is about 1/5 to 1/4 of the market capitalization of small stocks with institutional holdings. These ratios are remarkably stable across the years.

of-the-year return patterns. On aggregate, institutions are not liquidity providers; they actively trade, and their trades are related to the turn-of-the-year effect. In contrast to many previous studies, we show that the existence of tax-loss selling by individuals does not preclude the role of institutional trading in the turn-of-the-year effect. In essence, the results imply that the turn-of-the-year effect in loser small stocks is not only driven by tax-loss selling by individuals, but also by institutional trading.

4.2 Effects of Institutional Trading Intensity

In the preceding subsection, we only focus on aggregate institutional trading, but such an approach does not capture the varying objectives of different institutions as well as the trading dynamics. For example, for mutual funds, fund investors and the media typically focus more on the performance than the composition of the holdings of the funds. On the other hand, pension fund managers are more concerned about the stock holdings of their portfolios, because such information is typically closely evaluated by their sponsors. The incentives to window dress or risk adjust a portfolio may not be uniform across institutions. Moreover, the buying or selling of small stocks by institutions does not necessarily imply that their trading is large and active enough to cause any price pressure in the markets. The reason is that institutions have significantly lower holdings in small than large market-capitalization stocks. As Table 1 indicates, institutional holdings of small stocks are about 20 percent in 1998 and below 10 percent in 1986.

In light of the issues, we suggest an alternative approach to establish the direct link between the institutional buying/selling intensity and turn-of-the-year prices of small stocks. We employ a finer partitioning of the buy/sell classifications - the level of institutional buying/selling intensity - in order to capture the variation in returns around the turn of the year. The approach is as follows. Within the smallest size-formed quintile, we sort stocks independently based on the level of institutional buying and selling intensities. The level of institutional buying (selling) pressure of a stock is defined as the ratio of the number of shares of a stock bought (sold) by all institutions to the number of shares outstanding during

the last quarter of the year. The two ratios, to a certain degree, measure the price pressure generated by institutional trading activity in that quarter. The same sorting procedure is repeated for the trading activity in the first quarter of the year.

We recognize that the above approach cannot fully dichotomize buying pressure from selling pressure, as high institutional buying may coincide with high institutional selling. However, this approach is more suitable for our purpose than using net-buys, where a zero net-buy could be a result of significant buying and selling activities. Measuring the total sales and total purchases of each stock by all institutions separately can therefore capture the intensity of institutional buying or selling of a particular stock and hence its resulting impact on stock prices. In fact, the results below clearly yield distinctive return patterns for different levels of institutional trading intensity. We also recognize that our quarterly holdings data do not enable us to determine whether the window-dressing activity occurs at the end of the year and the risk-shifting strategy occurs at the beginning of the year. But our approach does not require that institutions engage in window-dressing activity only at the end of the year, or the risk-adjusting strategy only at the beginning of the year. Instead, we only require that the documented window-dressing activity in the fourth quarter is representative of institutional trading behavior at the end of the year, and similarly, the risk-adjusting activity in the first quarter is representative of trading behavior at the beginning of the year.¹⁴

Table 7 highlights the turn-of-the-year return patterns of extreme winner and extreme loser small stocks that institutions in aggregate have transacted during the last and first quarters of the year. Panel A reports the return patterns generated by the trading activity in the last quarter of the year, Panel B shows those in the first quarter of the year, and finally, Panel C emphasizes those conditioned jointly only on extreme trading intensities during the two adjacent quarters. The table reveals a number of interesting and contrasting return

¹⁴Using data on commercial papers, treasury bills and fund flow data, Musto (1997) shows that window dressing is more pronounced at the end of the year. His results suggest that there is no apparent reason for institutions to reshuffle their portfolio holdings until nearing reporting times. Thus, their study supports our claim that window-dressing activity by institutions is more likely to occur towards the end of the year.

differences.

We first look at the results in Panel A. For extreme loser stocks, December returns decrease, while January returns increase, monotonically from the low to the high selling-intensity quartile, producing an almost monotonic increasing differential returns in late December and early January, $\text{Diff}(J, D)$. We observe the same pattern in sales of extreme winner stocks, but a completely reversed pattern in purchases of extreme loser and winner small stocks. The results therefore suggest that when institutional selling pressure during the fourth quarter gets more intense, stock prices depreciate more in late December but also rebound more in early January. This turn-of-the-year effect is most pronounced in sales of extreme loser stocks from the two high selling-intensity groups, with $\text{Diff}(J, D)$ returns from 9.9 percent (t -ratio= 5.8) to 10.6 percent (t -ratio= 4.1).

On the other hand, active buying of loser small stocks by institutions during the last quarter of the year substantially weakens the January effect. This result is possibly due to the fact that institutions are providing liquidity in the market and hence reducing the selling pressure of the stocks. Stocks that have experienced large institutional buying pressure enjoy no significant price increase in early January; the $\text{Diff}(J, D)$ return of 0.3 percent is statistically insignificant. However, stocks in the lower three buying-intensity quartiles experience a stronger and statistically-significant January effect, with turn-of-the-year returns ranging from 4.8 percent (t -ratio= 3.9) to 11.3 percent (t -ratio= 6.0). As institutional buying pressure of extreme loser stocks heightens, the turn-of-the-year effect becomes small and statistically insignificant. In comparison, this effect is only marginally significant in the bottom two buying-intensity quartiles of winner stocks and is completely alleviated in the top two.

In Panel B, the return patterns generated from the trading activity in the first quarter of the year provide contrasting differences to those reported in Panel A. For loser stocks, the greater the institutional selling intensity in the first quarter of the year, the lower is the turn-of-the-year effect. Conversely, the higher their buying intensity, the larger is the turn-

of-the year effect. Interestingly, extreme loser stocks from the low selling-intensity and high buying-intensity groups produce similar return patterns, and garner statistically-significant turn-of-the-year returns of about 9.1 to 10.3 percent. Such similarities in return patterns are also evident in the low selling intensity and high buying-intensity groups of winner stocks. In contrast to the results in Panel A, extreme winner stocks in these two groups generate statistically-significant turn-of-the-year returns of about 3.2 to 4 percent.

Finally, Panel C provides a sharper contrast of the turn-of-the-year return differences by conditioning them jointly on institutional trading intensities in the fourth and first quarters of the year. In particular, it highlights return patterns from institutional selling in the fourth quarter ('Dec Selling') and institutional buying in the first quarter ('Jan Buying'). It is evident in the panel that any high Dec selling, especially accompanied by high Jan buying, would generate substantial price pressure on loser and winner small stocks around the turn of the year. Specifically, the trading pressure caused by both high Dec selling and high Jan buying of extreme loser small stocks yields the largest turn-of-the-year return of 12 percent (t -ratio= 3.6) and of extreme winner stocks is 3.6 percent (t -ratio= 2.6). On the contrary, when Dec selling and Jan buying of loser stocks are both low, we do not observe any turn-of-the-year effect.

In summary, institutional trading activity contributes significantly to the turn-of-the-year effect in small stocks. Even though institutions have no large holdings of small stocks, the concerted trading behavior, driven by similar incentives such as window dressing or risk adjusting, is able to affect the prices of these stocks. As we have stressed earlier, the evidence of tax-loss selling by individual investors, which is also supported by our results, does not nullify the effect of institutional trading activity. Clearly, institutional buying of small stocks towards the end of the year helps to reduce the turn-of-the-year effect, while institutional selling strengthens the effect. Similarly, institutional buying of winner small stocks at the beginning of the year also causes a significant, though weaker, January effect in winner stocks.

5 Conclusion

The well documented turn-of-the-year effect has generated considerable research in the finance literature. Several recent studies have provided evidence that the effect is attributed to tax-loss selling by individual investors. Such evidence, however, is drawn primarily from observing stock volume and return patterns generated around the turn of the year, and no explicit linkage can be established between turn-of-the-year return anomalies and the trading of individual investors vis-à-vis institutional trading. Nevertheless, what remains unclear is whether institutions have any impact on the dramatic price changes of small stocks around the turn of the year. In this paper, we use detailed institutional stock holdings data to investigate the trading activity of institutions around the turn of the year and find substantial evidence that institutional trading contributes significantly to the effect.

Results show that institutions tend to sell more extreme loser small stocks in the last quarter of the year, but buy more small stocks, both winners and losers, in the following quarter. These results provide evidence that institutions engage in window dressing towards the end of the year and adjusting the risk of their portfolios after the turn of the new calendar. We show that such institutional trading strategies are directly linked to the turn-of-the-year return patterns of small stocks. The findings suggest that institutions buying extreme loser small stocks helps to mitigate the turn-of-the-year effect, while selling intensifies the effect. Moreover, institutions buying more small stocks at the beginning of the year contributes to the observed January effect in winner stocks.

Even though institutional holding of small stocks is relatively low, the concerted trading behavior, driven by similar incentives such as window dressing or risk manipulating, significantly affects the prices of the stocks. Earlier studies on the turn-of-the-year effect have generally viewed the tax-loss selling hypothesis and window dressing hypothesis as two competing explanations. We, however, show that both individual and institutional investors are driving trading and price patterns of small stocks around the turn of the year.

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Table 1
Distribution of Institutional Stock Holdings

At the end of each September, NYSE, AMEX, and NASDAQ stocks are sorted into five size quintiles, according to the breakpoints of the market capitalization of NYSE stocks. The table shows the total number of stocks from each size quintile (Total), the proportion of the number of stocks in each quintile held by institutions (% Hold), and the fraction of the market value of stocks in each quintile owned by institutions (% Owned). The entire sample period is from January 1986 through December 1998.

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998
Smallest Stocks													
Total	3627	4018	3927	3802	3617	3431	3385	3640	4187	4150	4425	4730	4622
% Hold	77.1	77.3	80.8	82.3	82.6	84.6	88.4	90.8	91.4	91.3	91.0	91.8	94.4
% Owned	9.3	9.3	9.6	10.4	10.1	10.8	12.4	14.5	16.1	16.6	17.4	18.9	19.6
Quintile 2													
Total	826	837	832	771	810	877	971	982	937	1015	1072	987	952
% Hold	97.7	97.4	98.8	99.5	99.3	99.4	99.3	99.3	98.9	99.1	99.5	99.1	99.6
% Owned	25.1	26.1	26.8	30.6	29.4	31.7	32.4	37.2	38.8	39.9	40.6	44.2	46.0
Quintile 3													
Total	592	595	597	571	574	571	613	581	609	620	664	642	670
% Hold	98.8	97.6	99.0	99.8	100.0	99.8	99.7	99.8	100.0	99.8	99.5	99.5	99.6
% Owned	33.7	35.5	35.4	37.4	39.8	42.7	43.1	46.4	48.2	48.5	49.5	53.4	52.3
Quintile 4													
Total	440	437	426	417	417	429	444	475	495	530	520	510	511
% Hold	98.6	98.6	100.0	100.0	100.0	100.0	99.5	100.0	99.6	99.4	99.6	99.6	99.4
% Owned	41.5	43.1	44.3	45.1	45.5	49.1	48.8	52.0	53.5	54.0	55.7	58.0	59.2
Largest Stocks													
Total	373	378	373	361	364	374	393	406	421	431	459	449	441
% Hold	98.9	98.7	100.0	100.0	100.0	99.7	100.0	100.0	99.5	99.5	99.3	100.0	99.8
% Owned	48.5	49.5	50.4	51.3	52.0	52.7	54.6	56.9	57.0	57.4	57.7	61.3	61.4

Table 2
Turn-of-the-Year Return Patterns Across Size-Formed Portfolios

At the end of each September, NYSE, AMEX, and NASDAQ stocks are sorted into five size quintiles, according to the breakpoints of the market capitalization of NYSE stocks. All stocks are divided into winner and loser stocks, whose January to November returns are greater and less than zero, respectively. Within the winner/loser groups, stocks are further divided into two groups of approximately equal in number. Stocks with lower returns in the loser group are termed as extreme losers, and stocks with higher returns in the winner group are termed as extreme winners. The table presents the average returns on all the stocks within each size quintile for the last ten trading days of December (Dec), the first ten trading days of January (Jan), and their difference (Diff(J, D)) together with t-statistics in parentheses. It also reports the same for extreme winner and loser stocks.

	All Stocks			Extreme Losers			Extreme Winners		
	Dec	Jan	Diff(J, D)	Dec	Jan	Diff(J, D)	Dec	Jan	Diff(J, D)
Panel A: Full Sample Period: January 1986 - January 1999									
Smallest Stocks	0.014	0.051	0.037 (29.0)	0.010	0.082	0.072 (25.0)	0.023	0.035	0.012 (5.65)
Quintile 2	0.029	0.017	-0.012 (-7.27)	0.027	0.031	0.005 (1.03)	0.037	0.012	-0.025 (-9.11)
Quintile 3	0.033	0.010	-0.023 (-14.5)	0.028	0.019	-0.009 (-1.77)	0.041	0.008	-0.033 (-10.8)
Quintile 4	0.030	0.008	-0.022 (-14.4)	0.025	0.021	-0.005 (-0.79)	0.035	0.005	-0.030 (-9.03)
Largest Stocks	0.021	0.008	-0.013 (-9.84)	0.013	0.017	0.004 (0.62)	0.026	0.008	-0.018 (-6.12)

Table 2 (continued)

	All Stocks			Extreme Losers			Extreme Winners		
	Dec	Jan	Diff(J, D)	Dec	Jan	Diff(J, D)	Dec	Jan	Diff(J, D)
Panel B: December 1986 - January 1990									
Smallest Stocks	0.006	0.044	0.039 (20.32)	0.005	0.065	0.059 (14.9)	0.008	0.032	0.024 (7.07)
Quintile 2	0.013	0.025	0.012 (4.55)	0.013	0.031	0.018 (2.54)	0.015	0.024	0.009 (1.92)
Quintile 3	0.014	0.016	0.002 (0.62)	0.015	0.012	-0.003 (-0.39)	0.015	0.016	0.001 (0.16)
Quintile 4	0.013	0.017	0.004 (1.76)	0.015	0.016	0.001 (0.15)	0.008	0.020	0.012 (2.18)
Largest Stocks	0.004	0.013	0.009 (3.86)	-0.001	0.011	0.012 (0.98)	-0.001	0.016	0.017 (3.45)
Panel C: December 1990 - January 1994									
Smallest Stocks	0.023	0.068	0.046 (17.8)	0.023	0.091	0.068 (11.5)	0.028	0.066	0.038 (9.31)
Quintile 2	0.041	0.033	-0.008 (-2.50)	0.039	0.041	0.003 (0.27)	0.051	0.040	-0.012 (-2.45)
Quintile 3	0.048	0.022	-0.026 (-10.4)	0.042	0.025	-0.017 (-2.20)	0.064	0.032	-0.032 (-6.28)
Quintile 4	0.040	0.023	-0.018 (-8.12)	0.033	0.037	0.004 (0.49)	0.058	0.028	-0.030 (-5.48)
Largest Stocks	0.033	0.018	-0.014 (-7.20)	0.025	0.026	0.001 (0.08)	0.050	0.034	-0.016 (-2.56)
Panel D: December 1994 - January 1999									
Smallest Stocks	0.015	0.043	0.028 (12.6)	0.005	0.097	0.092 (16.7)	0.029	0.007	-0.022 (-6.71)
Quintile 2	0.034	-0.006	-0.040 (-14.8)	0.030	0.022	-0.008 (-0.99)	0.041	-0.024	-0.066 (-15.1)
Quintile 3	0.040	-0.009	-0.049 (-15.9)	0.034	0.024	-0.010 (-0.90)	0.044	-0.020	-0.065 (-11.9)
Quintile 4	0.038	-0.014	-0.052 (-18.7)	0.030	0.011	-0.019 (-1.63)	0.043	-0.023	-0.066 (-12.1)
Largest Stocks	0.029	-0.006	-0.035 (-15.7)	0.018	0.014	-0.004 (-0.26)	0.037	-0.008	-0.044 (-11.3)

Table 3
Institutional Trading Activity During the Last Quarter of the Year

The table shows time-series averages of equal- and value-weighted institutional sales and purchases of stocks from the top (extreme winners) and bottom (extreme losers) performance quartiles in the fourth quarter. Panel A reports results for all stocks and Panel B for small stocks. In Panel A, the performance quartiles are constructed as follows. All stocks are divided into winner and loser stocks, whose January to November returns are greater and less than zero, respectively. Within the winner/loser groups, stocks are further divided into two approximately equal-number groups. Stocks with lower returns in the loser group are termed as extreme losers, and stocks with higher returns in the winner group are termed as extreme winners. In Panel B, we employ the same approach to form performance quartiles, but within the smallest size quintile, which is constructed as follows. At the end of each September, NYSE, AMEX, and NASDAQ stocks are sorted into five size quintiles, according to the breakpoints of the market capitalization of NYSE stocks. Quintile 1 consists of small stocks. The ‘Sell’ (‘Buy’) ratio is defined as the ratio of quarter 4’s stock sales (purchases) in a performance group to quarter 3’s holdings of the same divided by the ratio of quarter 4’s total sales (purchases) to quarter 3’s total holdings. t -test is for the test of equality of means between ‘Sell’ and ‘Buy’ (Diff(Buy, Sell)). The sample period is from January 1986 to December 1998. t -statistics are in parentheses.

	Equal-weighted			Value-weighted		
	Buy	Sell	Diff(Buy, Sell)	Buy	Sell	Diff(Buy, Sell)
Panel A: ‘Buy’ and ‘Sell’ Ratios for All Stocks						
Extreme Losers	1.352	1.576	-0.226 (-16.1)	1.616	1.684	-0.068 (-6.20)
Extreme Winners	1.103	1.057	0.047 (4.24)	1.100	1.037	0.062 (9.00)
Panel B: ‘Buy’ and ‘Sell’ Ratios for Small Stocks						
Extreme Losers	1.060	1.274	-0.215 (-12.1)	1.096	1.181	-0.085 (-6.20)
Extreme Winners	1.122	0.968	0.167 (9.77)	1.176	1.134	0.045 (3.07)

Table 4
Institutional Trading of Small and Large Stocks
During the First Quarter of the Year

The table shows time-series of mean equal- and value-weighted institutional sales and purchases of small and large stocks in the first quarter. At the end of each September, NYSE, AMEX, and NASDAQ stocks are sorted into five size quintiles, according to the breakpoints of the market capitalization of NYSE stocks. The ‘Sell’ (‘Buy’) ratio is defined as the ratio of stock sales (purchases) in a size group during the first quarter of the year to the holdings of the same during the last quarter of the previous year divided by the ratio of the total sales (purchases) during the first quarter to the total holdings in the last quarter of the year. t -test is for the test of equality of means between ‘Sell’ and ‘Buy’ (Diff(Buy, Sell)), and is for the test of equality of means between smallest stocks and largest stocks (Diff(Smallest, Largest)). t -statistics are in parentheses. The sample period is from January 1986 to January 1999.

	Equal-weighted			Value-weighted		
	Buy	Sell	Diff(Buy, Sell)	Buy	Sell	Diff(Buy, Sell)
Smallest Stocks	1.496	1.275	0.221 (10.7)	1.983	1.570	0.412 (21.4)
Largest Stocks	0.958	1.007	-0.049 (-8.13)	0.893	0.930	-0.037 (-11.5)
Diff(Smallest, Largest)	0.539 (35.0)	0.267 (19.9)		1.090 (68.6)	0.640 (46.8)	

Table 5
Institutional Trading of Small Stocks, Sorted by Performance,
During the First Quarter of the Year

The table shows time-series averages of equal- and value-weighted institutional sales and purchases of small stocks from four performance quartiles in the first quarter during the sample period. The performance quartiles are constructed as follows. At the end of each September, NYSE, AMEX, and NASDAQ stocks are sorted into five size-formed quintiles, according to the breakpoints of the market capitalization of NYSE stocks. All stocks from the smallest-size quintile are divided into winner and loser stocks, whose January to November returns are greater and less than zero, respectively. Within the Winner/Loser groups, stocks are further divided into two equal-number groups. Stocks with lower returns in the loser group are termed as extreme losers, and stocks with higher returns in the winner group are termed as extreme winners. The ‘Sell’ (‘Buy’) ratio is defined as the ratio of quarter 1’s stock sales (purchases) in a size group to the previous quarter’s holdings of the same divided by the ratio of quarter 1’s total sales (purchases) to the previous quarter’s total holdings. *t*-tests for the ‘Sell’ and ‘Buy’ are on whether the ratios equal 1. *t*-tests for the difference between ‘Sell’ and ‘Buy’ (Diff(Buy, Sell)) are for the test of equality of means. The sample period is from January 1986 to December 1998. *t*-statistics are in parentheses.

	Equal-weighted			Value-weighted		
	Buy	Sell	Diff(Buy, Sell)	Buy	Sell	Diff(Buy, Sell)
Extreme Losers	1.017 (1.38)	1.141 (12.3)	-0.124 (-6.10)	1.165 (16.7)	1.102 (11.2)	0.066 (4.61)
Extreme Winners	1.114 (10.1)	1.000 (0.00)	0.114 (7.79)	1.167 (19.1)	1.148 (17.8)	0.015 (1.17)

Table 6

Aggregate Institutional Trading and Turn-of-the-Year (TOY) Returns on Small Stocks

The table presents time-series averages of returns during the last ten trading days of December (Dec), mean returns during the first ten trading days of January (Jan), and the mean differences between January and December returns (Diff(J,D)) for extreme winner/loser small stocks. At the end of each September, NYSE, AMEX, and NASDAQ stocks are sorted into five size quintiles, according to the breakpoints of the market capitalization of NYSE stocks. Quintile 1 consists of small stocks. These stocks are further divided into winner and loser stocks, whose January to November returns are greater and less than zero, respectively. Within the extreme winner/loser groups, stocks are, in turn, classified into three sub-groups based on aggregate institutional trading activity: stocks where institutions are net sellers (-1), stocks where institutions are net buyers (1), and stocks with no institutional holding and trading (0). Panel A contains results from the last quarter's trading activity, whereas Panel B contains those from the first quarter's. t -statistics for the difference between Net-Seller and Net-Buyer groups, Diff(-1,1), and the difference between January and December (Diff(J,D)) are in parentheses. The entire sample period is from January 1986 through January 1999.

Activity	Extreme Losers				Extreme Winners			
	Dec	Jan	Diff(J, D)	Obs.	Dec	Jan	Diff(J, D)	Obs.
Panel A: TOY Returns from the Last Quarter's Trading								
Net Seller (-1)	-0.013	0.099	0.104 (5.00)	8351	0.014	0.032	0.017 (1.52)	4242
No Holding (0)	0.000	0.098	0.096 (4.61)	2487	0.014	0.031	0.027 (2.25)	954
Net Buyer (1)	0.012	0.082	0.069 (4.06)	6035	0.029	0.033	0.004 (0.24)	6493
Diff(-1,1)	-0.025 (-2.02)	0.017 (1.56)	0.036 (3.22)		-0.015 (1.48)	-0.001 (-0.32)	0.014 (1.43)	
Panel B: TOY Returns from the First Quarter's Trading								
Net Seller (-1)	0.012	0.084	0.071 (3.90)	7632	0.020	0.027	0.007 (0.48)	5078
No Holding (0)	0.008	0.092	0.090 (4.93)	2340	0.003	0.028	0.025 (2.12)	898
Net Buyer (1)	0.020	0.098	0.082 (4.24)	6895	0.033	0.038	0.005 (0.38)	6805
Diff(-1,1)	-0.008 (-0.82)	-0.014 (-1.67)	-0.010 (-1.35)		-0.013 (1.84)	-0.011 (-1.63)	0.002 (0.46)	

Table 7

**Turn-of-the-Year (TOY) Returns on Small Stocks, Classified by
the Trading Intensity of Institutions**

The table presents time-series averages of returns on small stocks for the last ten trading days of December (Dec), the first ten trading days of January (Jan), and their difference (Diff(J, D)) based on institutional trading intensities. Panel A reports the mean returns from the fourth quarter's trading intensity, Panel B for the first quarter's trading intensity, and Panel C highlights those jointly on extreme trading intensities during the two quarters (i.e., Dec Selling and Jan Buying). At the end of each September, the five size quintiles are formed by grouping all NYSE, AMEX, and NASDAQ stocks into five quintiles according to the breakpoints of the market capitalization of NYSE stocks. Quintile 1 consists of small stocks. Selling (Buying) Intensity is defined as the ratio of total institutional sales (purchases) to shares outstanding for each stock. Stocks are sorted into four quartiles based on Buy and Sell Ratios. The differences in average returns (Diff(Low, High)) for low and high trading intensity stocks are also reported. All t -statistics are in parenthesis. The entire sample period is from January 1986 to January 1999.

	Extreme Losers			Extreme Winners		
	Dec	Jan	Diff(J, D)	Dec	Jan	Diff(J, D)
Panel A: TOY Returns from the Fourth Quarter's Trading Intensity						
Selling Intensity						
Low	0.025	0.070	0.044 (2.76)	0.041	0.023	-0.018 (-0.97)
Quartile 2	0.015	0.091	0.074 (4.46)	0.034	0.027	-0.007 (-0.51)
Quartile 3	0.007	0.114	0.106 (4.05)	0.025	0.037	0.012 (0.79)
High	-0.002	0.098	0.099 (5.83)	0.015	0.041	0.025 (1.80)
Diff(Low, High)	-0.028 (-2.12)	-0.002 (-1.21)	-0.055 (-2.36)	0.0253 (2.16)	-0.018 (-0.81)	-0.043 (-1.86)
Buying Intensity						
Low	0.006	0.121	0.113 (5.99)	0.014	0.040	0.026 (1.79)
Quartile 2	0.004	0.113	0.108 (5.10)	0.015	0.039	0.024 (1.84)
Quartile 3	0.019	0.064	0.046 (3.92)	0.030	0.033	0.003 (0.18)
High	0.022	0.048	0.025 (1.46)	0.044	0.023	-0.021 (-1.18)
Diff(Low, High)	-0.017 (-1.38)	0.074 (3.06)	0.089 (3.50)	-0.031 (-2.55)	0.0169 (0.82)	0.0467 (2.04)

Table 7 - cont'd
**Turn-of-the-Year (TOY) Returns on Small Stocks, Classified by
the Trading Intensity of Institutions**

	Extreme Losers			Extreme Winners		
	Dec	Jan	Diff(J, D)	Dec	Jan	Diff(J, D)
Panel B: TOY Returns from the First Quarter's Trading Intensity						
Selling Intensity						
Low	0.013	0.104	0.091 (6.52)	0.013	0.045	0.032 (2.36)
Quartile 2	0.011	0.111	0.099 (5.03)	0.020	0.038	0.018 (1.40)
Quartile 3	0.018	0.086	0.068 (2.81)	0.036	0.030	-0.006 (-0.38)
High	0.020	0.050	0.030 (2.29)	0.034	0.013	-0.031 (-2.13)
Diff(Low, High)	-0.007 (-0.60)	0.054 (2.76)	0.061 (3.21)	-0.0205 (-1.62)	0.044 (2.00)	0.053 (2.29)
Buying Intensity						
Low	0.026	0.057	0.031 (1.74)	0.039	0.034	-0.004 (-0.44)
Quartile 2	0.024	0.075	0.051 (3.71)	0.026	0.029	0.003 (0.22)
Quartile 3	0.010	0.094	0.084 (4.41)	0.018	0.039	0.021 (1.47)
High	0.006	0.108	0.103 (5.22)	0.016	0.046	0.040 (3.15)
Diff(Low, High)	0.020 (1.55)	-0.052 (-2.10)	-0.072 (-2.72)	0.033 (2.22)	-0.012 (-1.08)	0.044 (2.14)

Table 7 - cont'd
**Turn-of-the-Year (TOY) Returns on Small Stocks, Classified by
the Trading Intensity of Institutions**

	Extreme Losers			Extreme Winners		
	Dec	Jan	Diff(J, D)	Dec	Jan	Diff(J, D)
Panel C: TOY Returns Based Jointly on the Fourth and First Quarters' Trading Intensities						
Dec Selling/Jan Buying Intensities						
Low/Low	0.033	0.042	0.009 (0.51)	0.042	0.025	-0.018 (-1.42)
Low/High	0.019	0.107	0.088 (2.54)	0.020	0.035	0.013 (1.22)
High/Low	0.008	0.107	0.099 (3.71)	0.010	0.026	0.015 (1.92)
High/High	-0.008	0.113	0.120 (3.57)	0.027	0.063	0.036 (2.61)